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Miniaturized and wideband metasurface antenna sensor for breast tumor detection

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ABSTRACT

A wideband and compact metasurface (MS) antenna sensor is introduced for detecting breast tumors. First, the sensor comprises the multilayer MS and a coplanar waveguide (CPW) feeding. The CPW feeding excites the multilayer MS based on parallel plate capacitors, which can enhance the bandwidth and reduce the profile height. Next, the specific absorption rate analysis of the sensor is obtained to evaluate its capability for tumor detection. Then, a comparison analysis of the reflection coefficient was obtained incorporating tumors of different radii. A significant difference in reflection coefficients was observed between normal breast and breast models with tumors. Finally, the designed antenna sensor and the breast phantom are fabricated. The measured results of the sensor's gain, bandwidth, and radiation pattern show good agreement with the simulated results. The miniaturized sensor with an aperture dimension of $0.39\lambda_0 \times 0.51\lambda_0$ can be obtained. At the same time, the proposed sensor can obtain a wide bandwidth from 4.67 GHz to 9.07 GHz (64.0 %), a low profile of $0.07\lambda_0$, and a maximum gain of 6.9 dBi. Furthermore, the measured variations in reflection coefficients between normal and malignant phantoms further demonstrate the sensor's effectiveness for breast tumor detection.

1. Introduction

Breast cancer represents a major health challenge globally, which is the highest incidence and one of the leading causes of cancer-related deaths in women [1]. Its aggressive nature and potential for metastasis often lead to severe morbidity and mortality [2]. Therefore, it underscores the urgent need for early detection and effective treatment strategies. Several methods are employed for the early identification of breast cancer, including X-ray imaging, ultrasound, microwave sensors and Magnetic Resonance Imaging (MRI). Mammography is the most common screening tool, utilizing low-dose X-rays to identify abnormalities in breast tissue [3]. However, X-ray exposure can cause bodily harm and increase the risk of cancer development [4]. Ultrasound imaging is often employed as a complementary technique, particularly for distinguishing between solid masses and fluid-filled cysts. Ultrasound imaging for cancer detection is limited by its reduced sensitivity in distinguishing malignancies and challenges in imaging dense tissues [5]. The MRI has high sensitivity in detecting soft tissues of the breast [6]. These methods collectively contribute to improving the accuracy and timeliness of breast cancer diagnosis. However, these methods are expensive and can only be applied in the advanced stages of breast cancer [7].

In recent years, microwave sensors have been extensively studied for breast tumor detection, offering low cost, non-contact operation, high efficiency and early detection. For example, a patch antenna based on reflection coefficient for tumor detection was proposed in [8]. Patch antennas have been studied in tumor detection because of simple structure, high gain, and low cost, but patch antennas face the challenge of narrow bandwidth [9–12]. Similarly, the loop antenna in [13] and a cavity-backed antenna based on substrate-integrated waveguide in [14] were employed as microwave sensors for detecting breast tumors with limited bandwidths of less than 3 %. Ultra-wideband (UWB) antenna sensors have been employed in tumor detection owing to their wideband characteristics and transmission coefficient properties [15]. In [16], the eagle-shaped UWB sensor with a impedance bandwidth of 146.3 % is introduced for tumor detection. An UWB antenna based on a defected ground structure has compact dimensions and achieves a bandwidth of 69.1 % in [17]. Similarly, the UWB antenna with a bandwidth of 126.5 % has been proposed for distinguishing tumors and healthy biological tissues [18]. These UWB antennas are based on modified

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monopole antennas, which achieve wide bandwidth. However, the omnidirectional radiation pattern of the monopole antennas inherently results in a low gain and susceptibility to external interference [19]. To address the omnidirectional radiation of the monopole, a metamaterial surface was loaded above the monopole antenna to enhance the gain in [20], resulting in a high profile of 10.2 mm. In [21], a monopole antenna integrated with a partially reflective surface for breast tumor detection achieves directional radiation, but it exhibits a profile of 15 mm. An ultrawideband patch antenna integrated with an artificial magnetic conductor attains a peak gain of 8.1 dBi and a profile of 12.4 mm [22]. In addition, Vivaldi antennas have been widely employed for breast-tumor detection owing to their simultaneous high gain and broad bandwidth [23,24]. Nevertheless, because Vivaldi antennas are end-fire antennas, their height is inherently large. Therefore, the development of wideband, compact, and low-profile directional antennas is of significant research interest.

Metasurface (MS) antennas have been extensively studied due to wideband characteristics and broadside radiations [25–28]. However, the large aperture size of MS antennas limits their practical applications. To solve this problem, significant works have been studied on the miniaturization of wideband MS antennas. Adding capacitive loading is an effective approach to achieving wideband and miniaturized MS antenna [29–31]. For instance, the traditional MS loading patch-via-wall structure is designed in [30], featuring dimensions of $0.35\lambda_0 \times 0.35\lambda_0$ and a 43.1 % bandwidth. In [31], a MS antenna based on capacitive loading is proposed, which has a size of $0.34\lambda_0 \times 0.34\lambda_0$ and a bandwidth of 44.4 %. The second method is to change the shapes of the MS [32–34]. A wideband MS antenna was miniaturized using H-shaped unit cells, which have a bandwidth of 38.8 % [33]. In [34], a compact wideband MS antenna based on a inter-embedded metasurface configuration is presented with a aperture size of $0.43\lambda_0 \times 0.43\lambda_0$ and a bandwidth of 41.4 %. The third method reduces antenna apertures by using high-permittivity substrates. In [35], a miniaturized MS antenna realizes a size of $0.46\lambda_0 \times 0.46\lambda_0$ and a bandwidth of 27 % by using high-permittivity substrates. The fourth method is to use effective matching networks and feeder networks [36–38]. In [37], a printed broadband MS antenna with a coplanar waveguide (CPW) feeding is introduced, which has an impedance bandwidth of 67.3 %. However, this MS antenna fed by a CPW has a large aperture size. A wideband MS antenna excited by a bow-tie-shaped slot is presented, which has a bandwidth of 54.4 % and a thick profile of $0.13\lambda_0$ [38]. In addition, combining multiple resonant modes is a common approach to achieve wideband miniaturized MS antennas [39–42]. In [40], a compact antenna with a aperture size of $0.46\lambda_0 \times 0.49\lambda_0$ combining TM_{10} mode and TM_{20} mode is proposed. However, its impedance bandwidth is only 21.8 %. Similarly, a 25 % impedance bandwidth is realized for simultaneous excitation multiple modes [41]. The two-layer stacked patch MS realizes a compact antenna with a bandwidth of 54.9 % and a height of 5.6 mm [41]. The above works have achieved miniaturization and wideband MS antennas, which have the challenges of high profile, structural complexity, and limited bandwidth.

In this article, a compact, low-profile, and wideband sensor based on multilayer MS is designed. Compared with the conventional UWB antenna sensors, the designed sensor achieves wide bandwidth, high gain, and broadside radiation, simultaneously. To verify performances of the designed sensor, the specific absorption rate (SAR) analysis and the reflection coefficient results are provided for tumors of varying radii and positions. These results ensure the sensor's effectiveness in tumor detection. Finally, the impedance bandwidths, radiation patterns, and realized gains of the proposed sensor were measured experimentally. Good agreements between the simulated results and the measured results validate the design method. The innovations of this article are primarily divided into the following three aspects.

1) A novel and wideband sensor based on the parallel plate capacitor was first proposed for breast tumor detection.
2) This article achieves a compact, wideband, low-profile, broadside-radiation sensor simultaneously. The antenna sensor with an aperture dimension of $0.39\lambda_0 \times 0.51\lambda_0$, a maximum gain 6.9 dBi, and an impedance bandwidth of 64.0 % is proposed.
3) Compared to conventional UWB antennas with omnidirectional radiation patterns, the proposed sensor features a directional radiation pattern, offering higher gain and enhanced immunity to external electromagnetic interference.

2. Antenna senor design and principles

2.1. Antenna senor configuration

Fig. 1 illustrates the geometric configuration of the proposed sensor. The antenna sensor is composed of three layers of F4BM substrate with a relative permittivity of 3.5 and a loss tangent of 0.0042. The thicknesses of the top, middle, and bottom layers are 0.5 mm, 2 mm, and 2 mm, respectively. The top layer integrates a conventional 4 × 4 metallic patch array. The middle layer contains identical double-sided 4 × 4 metallic patches interconnected through vertical metal vias. The bottom layer is designed with a CPW feeding structure to simultaneously excite both the top and middle layers of the metasurface (MS) antenna. The antenna was numerically modeled and optimized using Ansys HFSS electromagnetic simulation software. The specific design parameters are as follows: $L = 44$ mm, $h = 4.5$ mm, $h_1 = 0.5$ mm, $h_2 = 2$ mm, $w_1 = 5.6$ mm, $l_1 = 7.8$ mm, $g_1 = 1$ mm, $s_1 = 0.7$ mm, $w_2 = 1.6$ mm, $l_2 = 6$ mm, $s_2 = 2.5$ mm, $g_2 = 0.8$ mm, $g_3 = 4.2$ mm, $l_3 = 21.5$ mm, $s_3 = 0.2$ mm, $w_3 = 3$ mm, $w_4 = 9.6$ mm, $s_4 = 1$ mm, $w_5 = 2$ mm, $s_5 = 2$ mm.

2.2. Design of the compact metasurface

Fig. 2(a) and (b) depict the conventional patch metasurface (designated as *MS-1*) and the proposed compact metasurface design (designated as *MS-2*), respectively. The conventional *MS-1* comprises metallic patches, a dielectric layer, and a ground. This architecture can be analytically modeled as a combination of inductance and capacitance in parallel [43], where L corresponds to the inductive reactance of the dielectric substrate, and C represents the capacitance induced by the

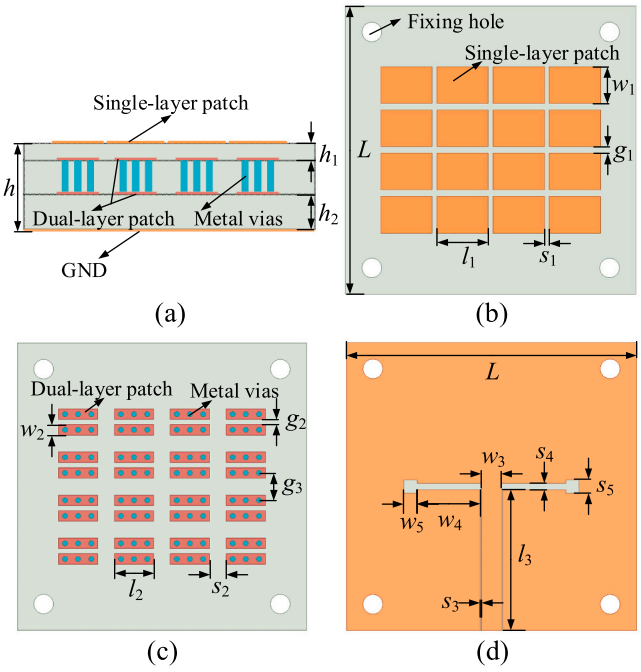


Fig. 1. Structure of the proposed sensor. (a) Side view. (b) Top layer. (c) Middle layer. (d) Bottom layer.

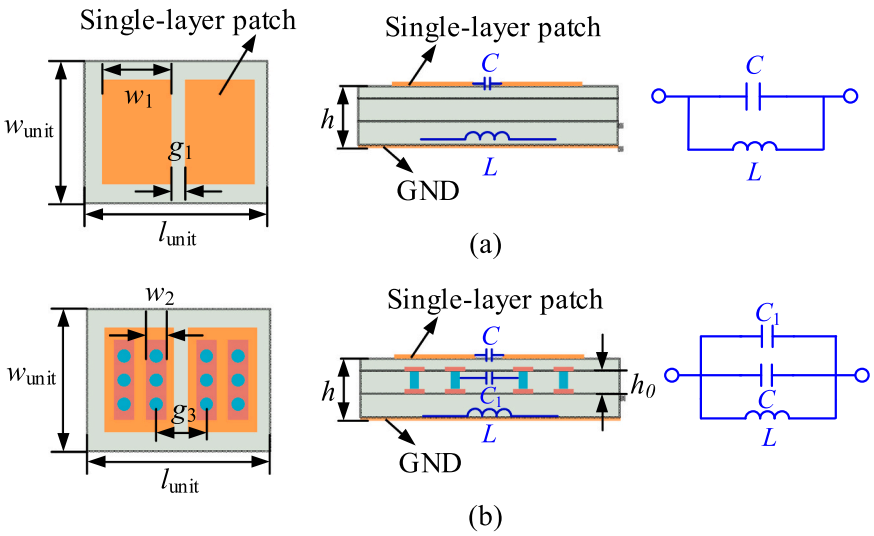


Fig. 2. Comparative structural and equivalent circuit analysis of MS configurations. (a) Conventional MS structure (denoted *MS-1*). (b) Proposed MS structure (denoted *MS-2*).

spacing between neighboring patches. The resonance frequency f_0 of this structure is determined by the following equation:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \tag{1}$$

Where the capacitance C and inductance L can be approximately given as:

$$C = \frac{w_1\varepsilon_0(1+\varepsilon_r)}{\pi}\cosh^{-1}\left(\frac{2w_1+g_1}{g_1}\right) \tag{2}$$

$$L = \mu h \tag{3}$$

where ε_r and μ denote the relative permittivity and the permeability of the dielectric substrate, respectively, while w_1 and g_1 represent the metallic patch width and inter-patch gap width between adjacent unit cells. As can be seen, the inductance L is only related to the thickness h and the material itself. Therefore, the method to achieve antenna miniaturization without increasing the profile mainly depends on increasing the capacitance C between the patches.

Building upon the above principle, we propose a compact multilayer MS architecture as can be seen from Fig. 2(b). The specific design parameters are as follows: $w_{unit} = 13.2$ mm, $l_{unit} = 17$ mm, $h = 4.5$ mm, $h_0 = 2$ mm, $w_1 = 5.6$ mm, $l_1 = 7.8$ mm, $g_1 = 1$ mm, $w_2 = 1.6$ mm, $l_2 = 6$ mm, $g_3 = 4.2$ mm. The proposed MS features an intermediate layer positioned between the radiating patches and ground plane. The middle layer consists of identical double-sided metal patches connected by metal vias. The middle layer is equivalent to adding a parallel plate capacitor to the conventional MS. The resultant equivalent capacitance can be defined as:

$$C_1 = \frac{\varepsilon_r S}{4\pi kd} \approx \frac{\varepsilon_r l_2 h_0}{4\pi k g_3} \tag{4}$$

Where S denotes the effective area of the capacitive plates, d represents the interplate separation distance, and ε_r corresponds to the relative permittivity of the dielectric medium. The resonant frequency of the proposed MS is shown in Eq. (5). As can be seen, the parallel plate capacitor can significantly increase the capacitance of the resonant circuit, thereby greatly reducing the resonant frequency f_1.

$$f_1 = \frac{1}{2\pi\sqrt{L(C+C_1)}} \tag{5}$$

Fig. 3 compares the numerical simulations of f_1 from Eq. (5) with

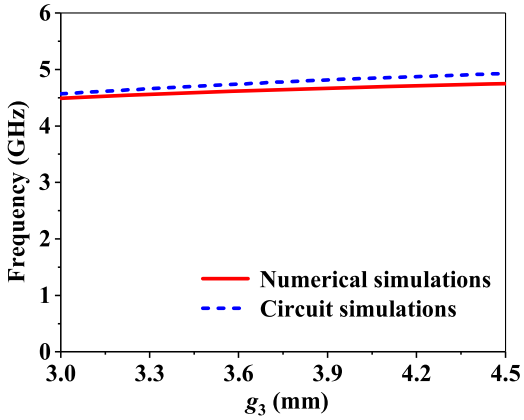


Fig. 3. Numerical and circuit-simulation results under different g_3 values.

circuit simulations. It can be seen that the resonant frequency of the proposed MS decreases monotonically with g_3.The numerical results align closely with the circuit results. This agreement confirms the accuracy of Eq. (4) and Eq. (5). These equations enable rapid estimation of the resonance frequency and shorten the antenna design cycle.

In order to compare the advantages of the proposed MS with the conventional MS in terms of size reduction. The structure of the proposed MS and conventional MS is shown in Fig. 2. The reflection phases between the conventional MS and the proposed MS was conducted under the same aperture, as illustrated in Fig. 4. The reflection phase of 0° indicates the resonant frequency. It can be seen that the resonance frequency f_1 of the proposed MS is significantly shifted to lower frequencies compared with the conventional MS. The proposed MS achieves a 28 % reduction in resonant frequency relative to the conventional design. This significant frequency shift confirms the efficacy of the proposed methodology in advancing structural miniaturization.

2.3. Design of the wideband antenna sensor

Following the development of a compact MS, a wideband, low-profile MS antenna utilizing the CPW feeding is proposed as illustrated in Fig. 1. Compared to conventional microstrip lines and aperture-coupled slot feeding methods, the CPW feed mechanism enhances

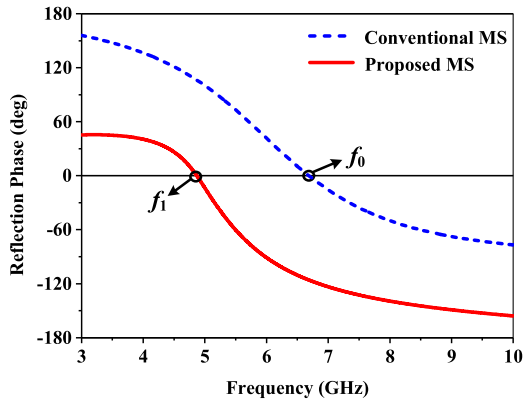


Fig. 4. Reflection phase with different MS structures under the same aperture.

bandwidth while reducing antenna thickness [44,45]. Fig. 5 compares the reflection coefficients of the proposed CPW-fed MS antenna with different feeding structures. A comprehensive analysis of the antenna's key performance characteristics is summarized in Table 1. The proposed sensor achieves a compromise between wideband functionality and compact dimensions. Moreover, the proposed *MS-2* exhibits a lower resonant frequency and enhanced impedance bandwidth compared to the conventional *MS-1*. The *MS-2* with the CPW feeding increases the fractional bandwidth from 40.9 % to 67.0 % compared to *MS-2* with aperture-coupled feeding. The CPW feeding not only reduces the antenna profile height but also improves the impedance-matching characteristics.

Based on the above analysis, the design methodology for the wideband MS antenna sensor can be systematically outlined in three stages:

1) First, after determining the resonant frequency and dielectric substrate thickness, the geometric parameters of the *MS-2* unit cell can be calculated based on Eqs. (2–5).
2) Then, the *MS-2* structure is modeled and optimized using the software Ansys HFSS. By adjusting the geometry sizes of the *MS-2* model, the resonance frequency can be optimized to the desired value.
3) Finally, the proposed *MS-2* structure is excited using the proposed CPW feeding. Meanwhile, optimizing the parameters of CPW feeding achieves wideband impedance matching.

3. Breast tumor detection

3.1. Beast model configuration

Fig. 6 illustrates the geometry of the hemispherical breast model,

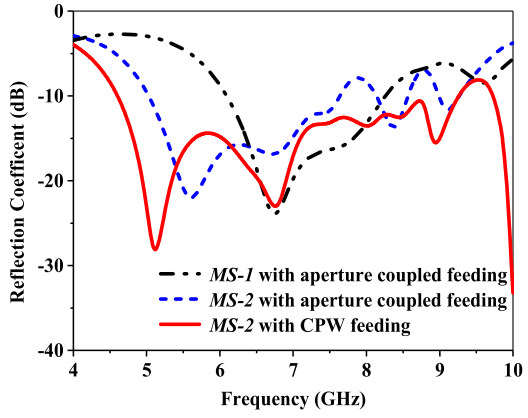


Fig. 5. Reflection coefficients of the conventional *MS-1* and the proposed *MS-2* with different feeding structures.

Table 1
Performances with Different MS and Feeding Structures.

Type	Bandwidth	Thickness (mm)	Aperture (mm × mm)
MS−1 with aperture-coupled feeding	30.0 %	5	32.6 × 22.6
MS−2 with aperture-coupled feeding	40.9 %	5	32.6 × 22.6
MS−2 with CPW feeding	67.0 %	4.5	32.6 × 22.6

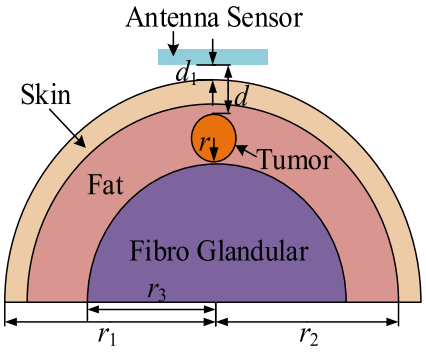


Fig. 6. Hemispherical breast model.

which comprises four layers: skin, fat, fibro glandular tissue, and tumor. The skin layer with a 5 mm thickness exhibits a relative permittivity of 36 and an electrical conductivity of 4 S/m. The fatty tissue with a thickness of 15 mm possesses dielectric characteristics of $\varepsilon_r = 9$ and $\sigma = 0.4$ S/m. The fibro glandular tissue with a 30 mm thickness demonstrates dielectric properties of $\varepsilon_r = 13$ and $\sigma = 0.45$ S/m. The tumor is modeled with a dielectric properties of $\varepsilon_r = 50$ and $\sigma = 4$ S/m. The relative permittivity and conductivity values for each layer were derived from [46]. The topology of the model was designed using Ansys HFSS software. The breast model has the following key design parameters: $d = 10.5$ mm, $d_1 = 1$ mm, $h_2 = 2$ mm, $r_1 = 50$ mm, $r_2 = 45$ mm, $r_3 = 30$ mm, $r = 5$ mm.

3.2. Specific absorption rate analysis

Specific Absorption Rate (SAR) refers to the absorption rate of electromagnetic energy by biological tissues under radio frequency irradiation. Mathematically, SAR can be represented as Eq. (6). Here, σ represents the electrical conductivity, $|E|$ represents the magnitude of the electric field, and ρ denotes the mass density.

$$SAR_{Local} = \frac{\sigma |E|^2}{\rho} \tag{6}$$

In breast cancer, the absorption of electromagnetic radiation by tumors differs significantly from that of healthy tissues. This distinction enables the detection of tumors and their precise locations based on variations in the SAR between healthy tissues and tumors. Fig. 7 illustrates the hemispherical breast model without tumors, along with the corresponding SAR values for the skin, fat, and fibro glandular tissue. As observed, the maximum local SAR values for the skin, fat, and fibro glandular tissue are 254.4 W/kg, 13.2 W/kg, and 3.8 W/kg, respectively. Fig. 8 depicts the hemispherical breast model with a 5 mm radius tumor and its SAR value for skin, fat, fibro glandular tissue, and tumor. Here, the maximum local SAR values are 241.7 W/kg, 18.8 W/kg, 3.1 W/kg, and 43.0 W/kg, respectively. The tumor is located 10.5 mm away from the antenna sensor. Notably, the SAR values for the skin, fat, and fibro glandular tissue in the tumor-embedded model exhibit significant variations compared to the tumor-free model. These variations can be effectively utilized for tumor detection. Specifically, the SAR in

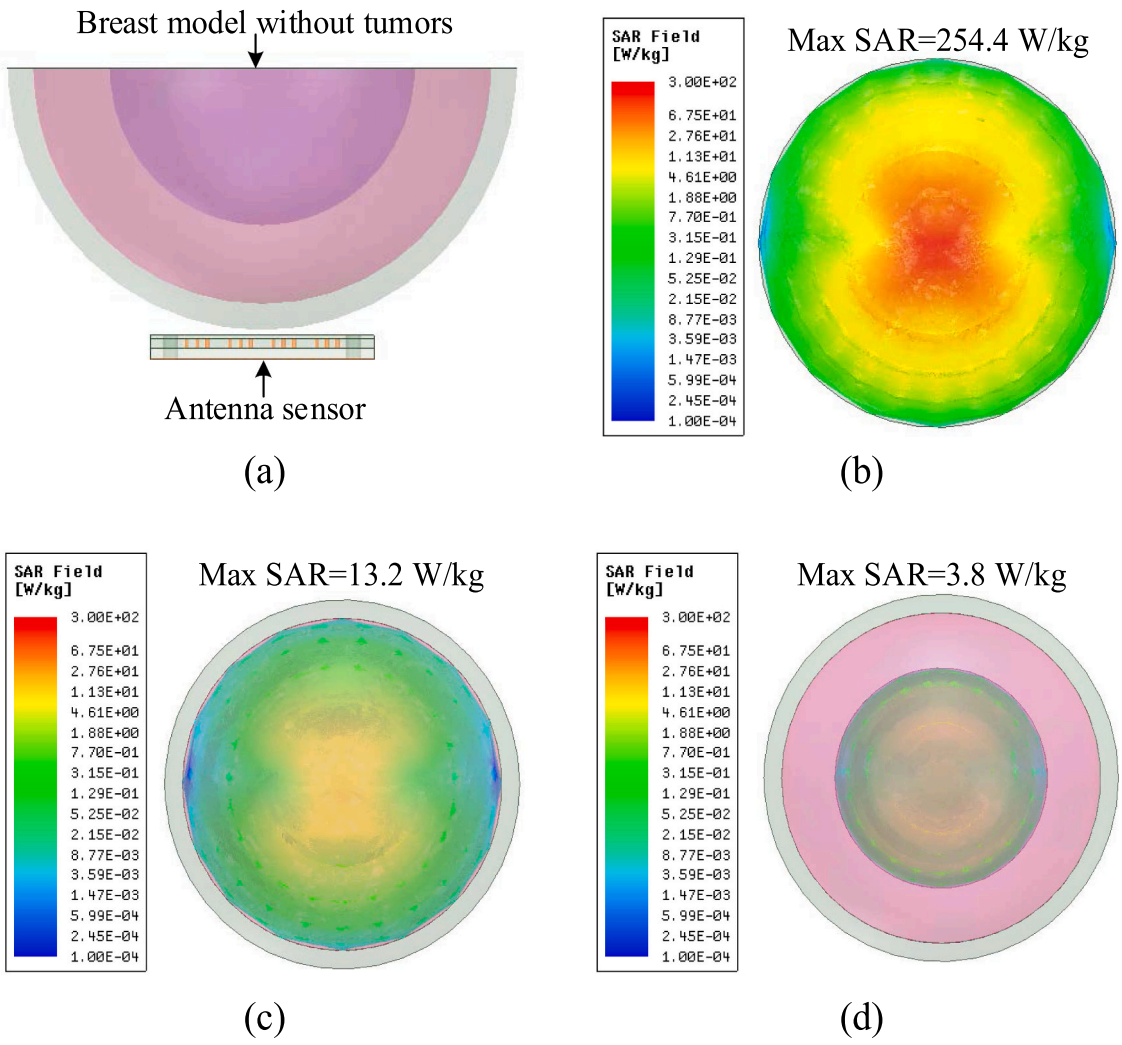


Fig. 7. Hemispherical breast model without tumors and its SAR value for skin, fat, and fibro glandular tissue. (a) Antenna sensor and breast model without tumors. (b) Local SAR value for skin. (c) Local SAR value for fat. (d) Local SAR value for fibro glandular tissue.

the fatty tissue is markedly higher in the tumor-embedded model than in the tumor-free model, primarily due to the tumor's presence within the fatty layer, which enhances energy absorption and elevates the SAR.

To further study the influence of tumor radius and SAR values, Table 2 provides the SAR values for tumors with radii ranging from 0 mm to 10 mm. As can be seen, the SAR value of the skin with a tumor is observed to be lower than that of the skin without a tumor. Additionally, as the tumor radius increases, the SAR value of the tumor correspondingly increases. Similarly, Table 3 shows the SAR values when the distance between the antenna sensor and the tumor ranges from 10.5 mm to 21.5 mm. It can be observed that when the tumor is located in the fat layer (d=10.5 mm), the SAR of the fat with the tumor is higher than that of the fat without the tumor. Similarly, when the tumor is located in the fibro glandular layer (d=21.5 mm), SAR values of the fibro glandular tissue with the tumor are higher than that of the fibro glandular tissue without the tumor. Furthermore, when the tumor is simultaneously present in both the fat and fibro glandular layers (d=16 mm), the SAR values for both tissues with the tumor are higher than those without the tumor. The above results demonstrate the relationship between tumor size, its distance from the antenna, and SAR, which is of critical importance for tumor detection. It is crucial to stress that, although these SAR values surpass the Federal Communications Commission (FCC) guideline limit of 1.6 W/kg per 1 g, they can be made to fall within the safe range through a reduction in the input power.

3.3. *Reflection coefficient of antenna sensor for tumors with different radii*

To validate the effectiveness of the proposed antenna sensor in detecting breast tumors, Fig. 9 depicts the simulated reflection coefficients for tumors with radii of 0 mm, 3 mm, 5 mm, and 10 mm. The breast model is shown in Fig. 6. which is composed of four concentric layers, the skin layer(ε_r =36, σ=4 S/m), the fat layer (ε_r =9, σ=0.4 S/m), the fibro glandular tissue (ε_r =13, σ=0.45 S/m), and the tumor (ε_r =50, σ=4 S/m). The observed variation in the sensor's reflection coefficient stems from the distinct electromagnetic properties of tumorous versus healthy tissues. Notably, the resonant frequency of the model incorporating a tumor shifts to a higher frequency relative to the tumor-free model. Moreover, as the tumor radius increases, the resonant frequency further shifts toward higher frequencies. Tumors with different radii exhibit significantly greater differences in high-frequency responses within the bandwidth compared to low-frequency responses. This is because the wavelengths corresponding to high frequencies are shorter, making them more sensitive to tumor size. It is also an advantage of wideband antenna sensors over narrow-band antenna sensors. These characteristics facilitate the detection of tumors and the estimation of their size.

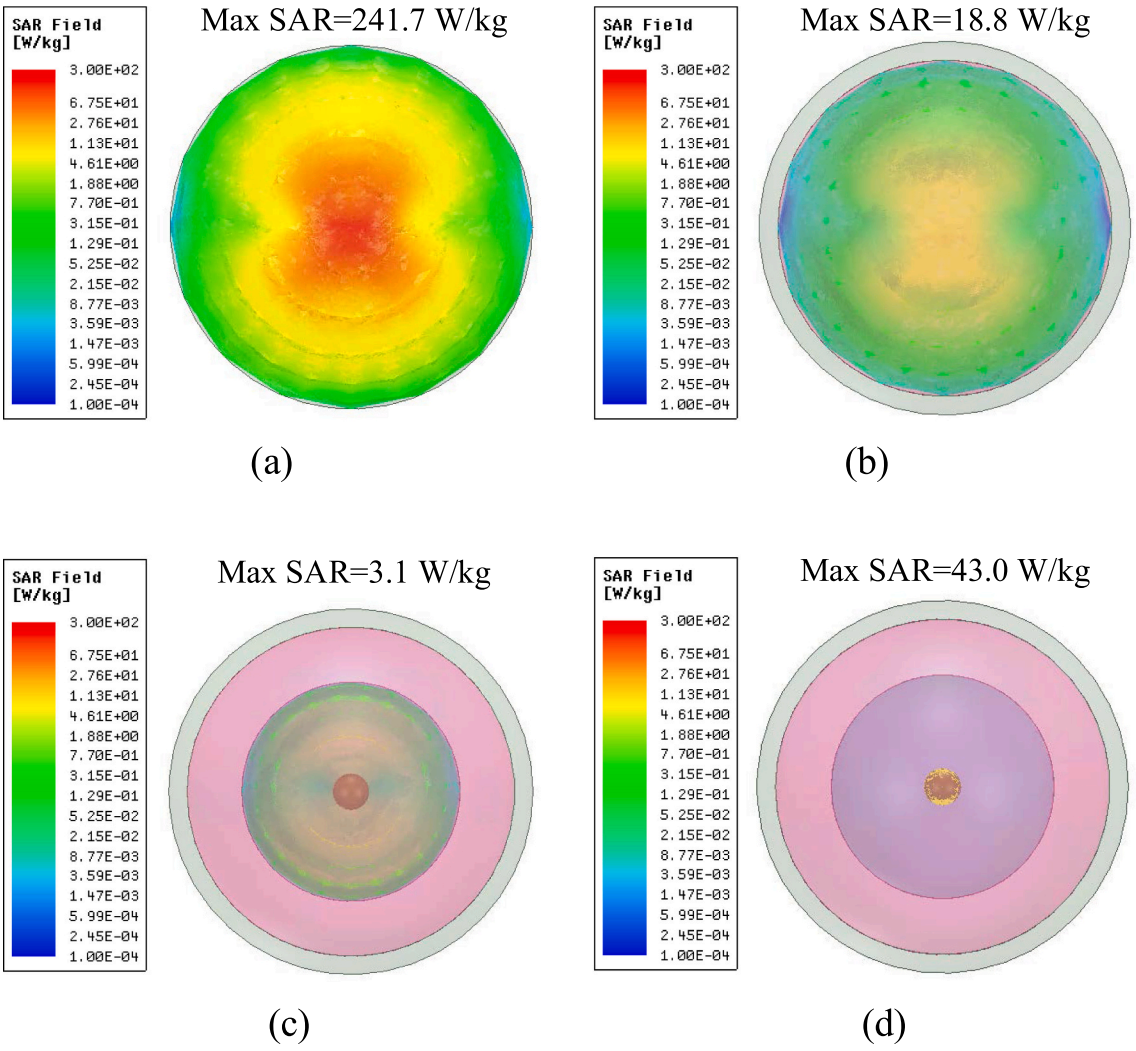


Fig. 8. Hemispherical breast model with a 5 mm radius tumor and its SAR value for skin, fat, fibro glandular tissue, and tumor. (a) Local SAR value for skin. (b) Local SAR value for fat. (c) Local SAR value for fibro glandular tissue. (d) Local SAR value for tumor.

Table 2
SAR Analysis for Tumors with Various Radii (Sensor to Tumor Distance of 16 mm, Input Power = 1 W).

Tumor radius (mm)	Skin SAR (W/kg)	Fatty SAR (W/kg)	Fibro Glandular SAR (W/kg)	Tumor SAR (W/kg)
No tumor	254.4	13.2	3.8	/
3	242.7	17.4	8.9	19.2
5	244.2	13.7	5.9	19.9
10	227.6	13.4	4.9	21.0

Table 3
SAR Analysis for Tumors with Various Distances (Tumor Radii of 5 mm, Input Power = 1 W).

Distance *d* (mm)	Skin SAR (W/kg)	Fatty SAR (W/kg)	Fibro Glandular SAR (W/kg)	Tumor SAR (W/kg)
No tumor	254.4	13.2	3.8	/
10.5	241.7	18.8	3.1	43.0
16	244.2	13.7	5.9	19.9
21.5	247.8	13.0	9.6	17.7

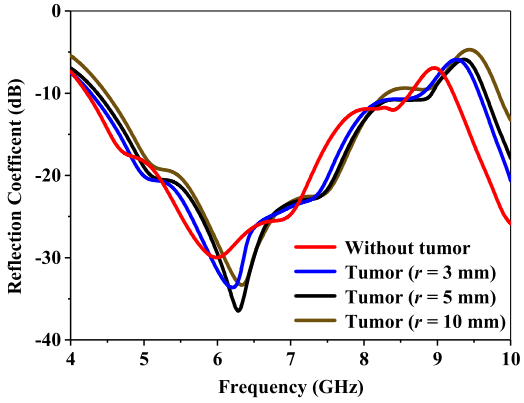


Fig. 9. Simulated reflection coefficients for tumors with different radii.

4. Fabrication and measurement results

4.1. Antenna sensor fabrication and measurements

To verify the design, the proposed antenna sensor was fabricated and then subjected to measurement. Fig. 10 presents the fabricated prototype of the antenna sensor. Given the wide bandwidth of the antenna

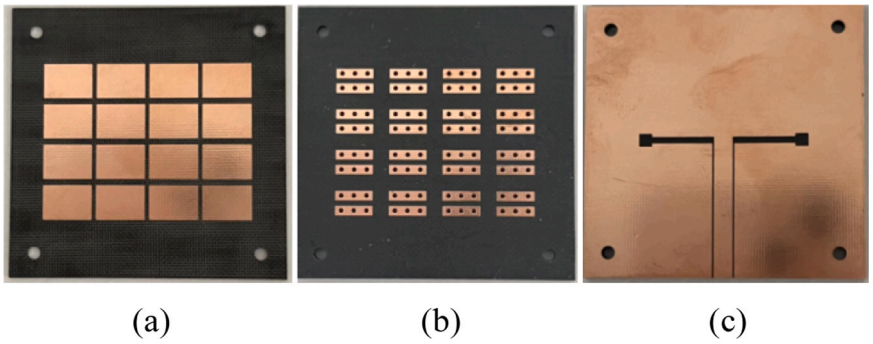


Fig. 10. The fabricated antenna prototype. (a) Top layer. (b) Middle layer. (c) Bottom layer.

sensor, its far-field performances were measured in both the multi-probe anechoic chamber and the near-field anechoic chamber, as depicted in Fig. 11. The radiation patterns and realized gain of the proposed antenna within the frequency range from 4 GHz to 6 GHz were measured using the multi-probe anechoic chamber. Likewise, the radiation patterns and realized gain of the proposed antenna within the frequency range from 6 GHz to 10 GHz were measured using the multi-probe anechoic chamber.

The simulated and measured reflection coefficients are depicted in Fig. 13. As can be seen, the measured bandwidth where the reflection coefficients are below −10 dB ranges from 4.67 GHz to 9.07 GHz (64.0 %). In terms of simulation, the bandwidth with reflection coefficients under −10 dB spans from 4.61 GHz to 9.25 GHz (67.0 %). Overall, there is a good agreement between the measured and simulated results.

Fig. 13 presents the simulated and measured radiation patterns of the antenna sensor at 5.73, 6.98, 8.51, and 9.02 GHz. It is evident that the measured radiation patterns closely align with the simulated ones. Additionally, the cross-polarization in the maximum radiation direction remains below −20 dB. Fig. 14 illustrates the simulated and measured realized gain of the antenna sensor. The measured and simulated peak gain of the antenna sensor is 6.9 dBi and 7.2 dBi respectively, showing a good agreement between the two results. However, it should be noted that the proposed antenna exhibits relatively low gain at high frequencies. This limitation could be significantly mitigated by introducing an appropriate height of air layer between the CPW feeding and the designed microstrip (MS). As shown in Fig. 15, it can be seen that when the height of the air layer is 2 mm, the whole band gain is significantly enhanced. Nevertheless, in the context of breast tumor detection using the designed antenna sensor, profile height and stability are of paramount importance. Therefore, the air layer is not utilized to enhance the antenna gain. In summary, the proposed antenna features an aperture of $0.39\lambda_0 \times 0.51\lambda_0$, a wide impedance bandwidth from 4.67 GHz to 9.07 GHz (64.0 %), a low profile of $0.07\lambda_0$ and a peak gain of 6.9 dBi.

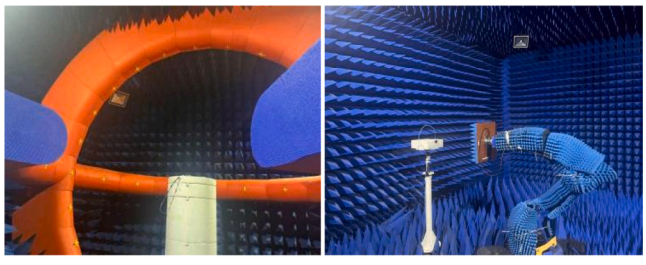


Fig. 11. The measurement setup. (a) Multi-probe anechoic chamber setup. (b) Near-field anechoic chamber setup.

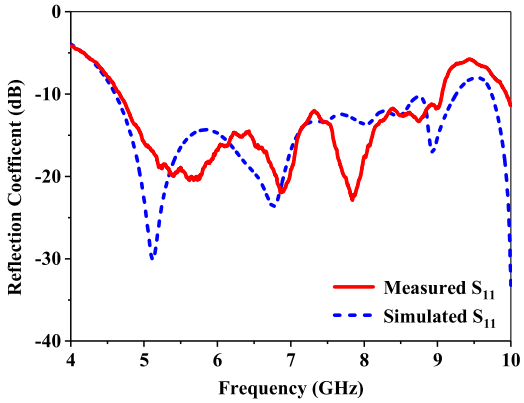


Fig. 12. Measured and simulated reflection coefficients of the proposed antenna.

4.2. Phantom fabrication and measurements

The hemispherical breast model was fabricated using a mixture of oil, gelatin, water, and sodium chloride according to the method described in [15]. As shown in Table 4, the dielectric constant and the electrical conductivity of each tissue component in the breast model were adjusted by varying the proportions of oil, gelatin, water, and sodium chloride. The manufacturing process of the breast model consists of four stages: First, the mixture solution was prepared using an electronic balance and measuring cylinder according to Table 4. Second, the mixture solution was heated and stirred at 85 °C for 15 min to achieve homogenization, as illustrated in Fig. 16(a). Third, the heated and mixed solution was poured into a 3D-printed hemispherical mold and allowed to cool and solidify. Fourth, tumor models (radii of 3 mm, 5 mm, and 10 mm), fibro glandular tissue, fatty tissue, and skin tissue models were individually fabricated, and finally assembled into the hemispherical breast model. To facilitate assembly, all tumors of different radii were positioned 21 mm away from the sensor. Additionally, 1 g of methylene blue dye was added to the tumor models to distinguish them from other tissues. The completed breast model is shown in Fig. 16(b).

The measured reflection coefficients for tumor-free and tumor-embedded models with radii of 3 mm, 5 mm, and 10 mm are shown in Fig. 17. The antenna exhibits a −10 dB impedance bandwidth of 4.35–7.79 GHz without tumor in breast phantom. Introducing a 3-mm-radius tumor shifts this band upward to 4.38–7.87 GHz. Further enlarging the tumor to 5 mm yields a bandwidth of 4.62–7.99 GHz, while a 10-mm-radius tumor extends a final band of 4.87–8.19 GHz. It is evident that the resonance frequency of the tumor-embedded models shifts to higher frequencies in comparison to the tumor-free model. Moreover, as the tumor radius increases, the difference in reflection coefficients between the tumor-embedded and tumor-free models becomes more significant. These characteristics are conducive to

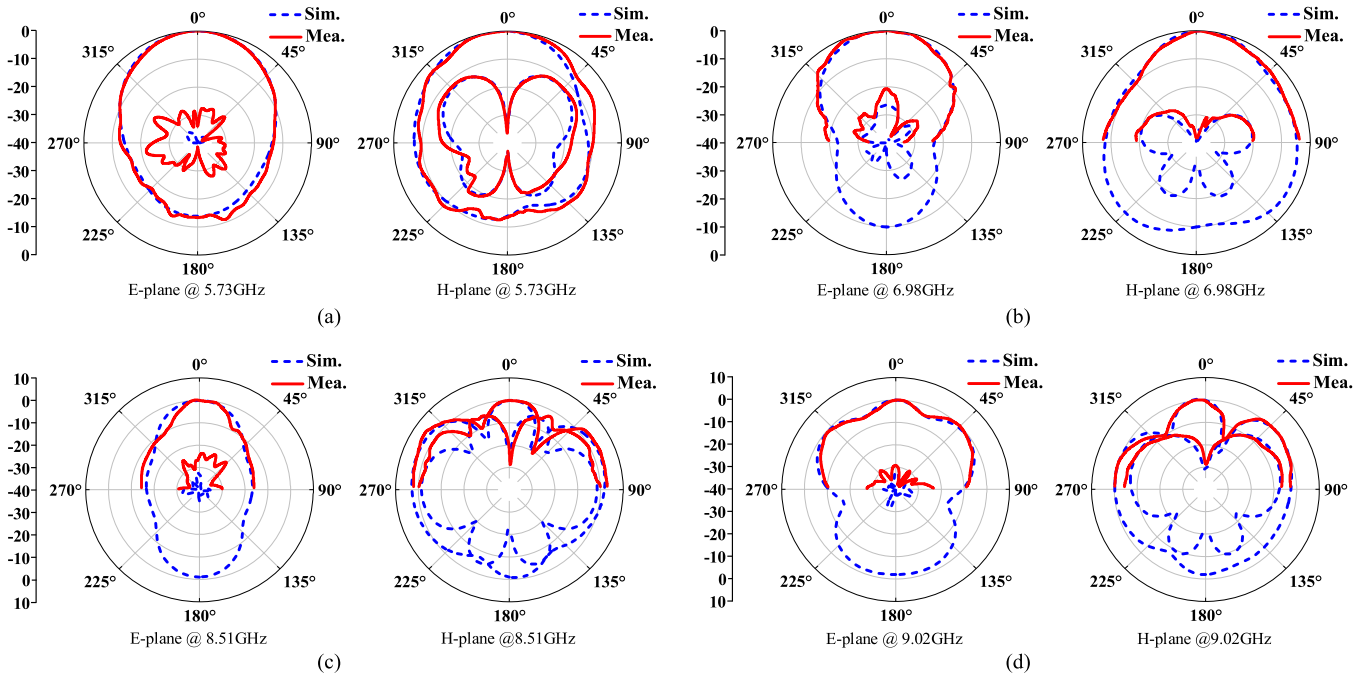


Fig. 13. Normalized radiation patterns of the antenna sensor. (a) 5.73 GHz. (b) 6.98 GHz. (c) 8.51 GHz. (d) 9.02 GHz.

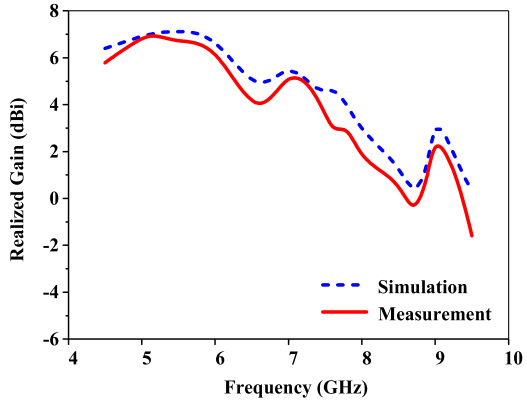


Fig. 14. Measured and simulated gain of the proposed antenna sensor.

facilitating tumor detection and size estimation. The measured reflection coefficients exhibit good agreement with the simulated results, thereby confirming the efficacy of the proposed antenna sensor in detecting breast tumors.

Fig. 18 presents the measured reflection coefficients for a 5 mm-radius tumor placed at various positions. When the tumor is located directly in front of the sensor (Locations 1 and 5), a pronounced frequency shift relative to the tumor-free case is observed, enabling reliable detection. Conversely, at large oblique angles (position 3), the reflection coefficient closely overlaps the tumor-free case, rendering the tumor almost undetectable. Therefore, in practical tumor detection, a large number of antenna sensors need to be arranged around the breast or the relative rotation method to achieve tumor detection.

Table 4
Compositions of phantom materials.

Tissue	Gelatin (gram)	Water (mL)	Oil (mL)	sodium chloride (gram)
Tumor	25	75	14	2.5
Fibro Glandular	50	160	42	0
Fatty	20	40	154	0
Skin	43	150	47	0

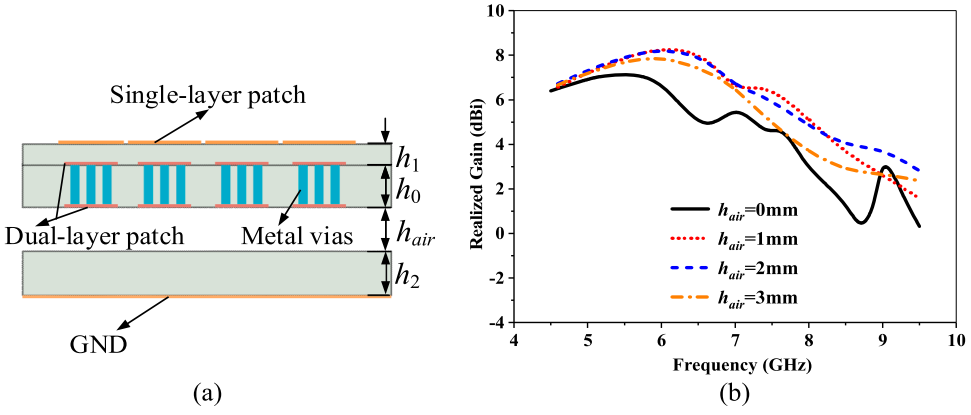


Fig. 15. Realized gain with different air layer heights. (a) Antenna structure with the air layer. (b) Parametric simulation analysis.

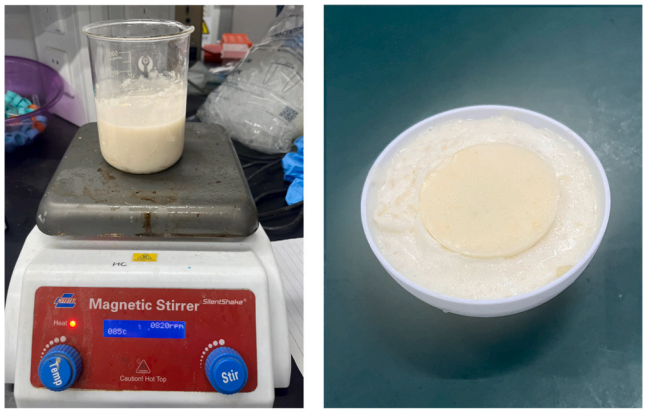


Fig. 16. Fabrication process of the breast phantom. (a) The mixture solution heated in a beaker. (b) The breast phantom after solidification.

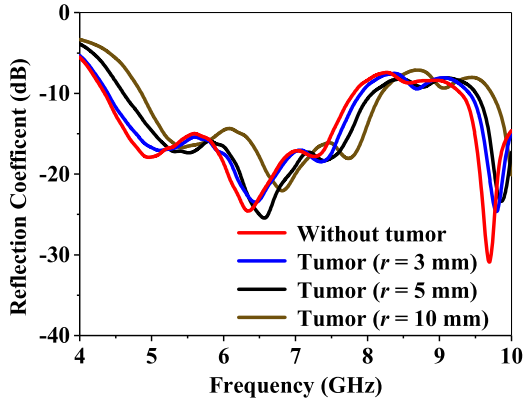


Fig. 17. Measured reflection coefficients for tumors with radii of 0 mm, 3 mm, 5 mm, and 10 mm.

To understand the performances of the antenna sensor in aspects of tumor detection, bandwidth, size, and gain improvements, a comparative analysis between the designed sensor and other relevant antennas was carried out listed in Table 5. Here, λ_0 represents the wavelength in free space at the lower frequency of the impedance bandwidth. In comparison, the sensors presented in [16], and [17] exhibit wider bandwidth and smaller dimensions than the designed sensor. However, these sensors feature omnidirectional radiation patterns, which inherently result in lower gain and increased susceptibility to external electromagnetic interference. References [20] and [23] present wideband, high gain, and compact antennas. However, their profile heights substantially exceed that of the proposed design. The proposed sensor effectively combines wide bandwidth, small size, low profile, and high gain. These excellent properties enable the designed antenna sensor to be applied in breast tumor detection.

5. Conclusion

A compact and wideband antenna sensor based on the multilayer MS is first proposed in this article. To analyze the relationship between the geometry dimensions and the resonant frequency of the multilayer MS, an equivalent circuit model is developed. Following the design of the compact MS, a simple CPW feeding is employed to excite the structure, which enhances the bandwidth while reducing the profile height. The proposed antenna sensor exhibits an antenna size of $0.68\lambda_0 \times 0.68\lambda_0$ and achieves a wide operating bandwidth ranging from 4.67 to 9.07 GHz. Additionally, it demonstrates a maximum realized gain of 6.9 dBi and a cross-polarization level below −20 dB. To validate its effectiveness in tumor detection, a SAR analysis is conducted for both healthy and tumor tissues. The results reveal significant differences in SAR, which can distinguish healthy and tumor tissues. Furthermore, tumors with different radii exhibit significantly greater differences in reflection coefficients. Finally, the proposed sensor and the breast phantom are fabricated and experimentally measured. Measurement results show that the proposed sensor is able to effectively detect tumors with a radius of more than 3 mm. The designed sensor simultaneously achieves simple structure, wide bandwidth, low profile, compact size and high gain, which has great potential for breast tumor detection.

Author statement

We would like to submit the enclosed manuscript entitled "**Miniaturized and Wideband Metasurface Antenna Sensor for Breast Tumor Detection**", which we wish to be considered for publication in **Sensors and Actuators A: Physical**.

A wideband and compact metasurface (MS) antenna sensor is introduced for detecting breast tumors. First, the sensor comprises the multilayer MS and a coplanar waveguide (CPW) feeding. The CPW feeding excites the multilayer MS based on parallel plate capacitors, which can enhance the bandwidth and reduce the profile height. Next, the specific absorption rate analysis of the sensor is obtained to evaluate its capability for tumor detection. Then, a comparison analysis of the reflection coefficient was obtained

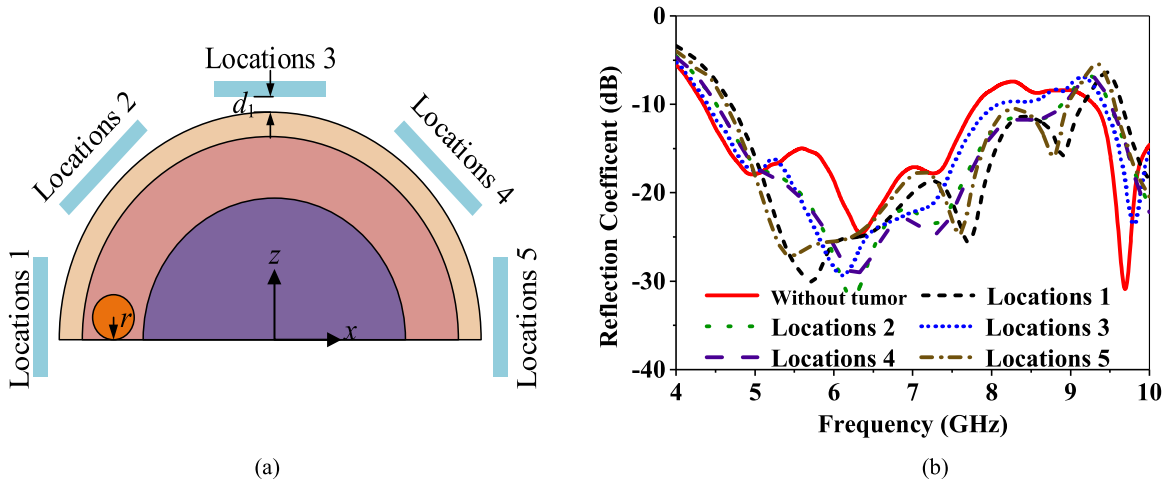


Fig. 18. Measured reflection coefficients at different locations. (a) Sensor structure locations. (b) Measured reflection coefficients.

Table 5
Comparisons with Other Relevant Antennas.

Ref.	Center frequency (GHz)	Bandwidth	Peak gain (dBi)	Antenna size	Antenna height	Breast tumor detection	Radiation pattern
[13]	1.108	< 1 %	N.M	0.46 λ_0× 0.35 λ_0	0.02 λ_0	Yes	Omnidirectional
[14]	2.425	2.1 %	3.5	0.38 λ_0× 0.30 λ_0	N.M	Yes	Directional
[16]	10.4	146.3 %	5.49	0.24 λ_0× 0.20 λ_0	0.02 λ_0	Yes	Omnidirectional
[17]	9.29	69.1 %	4.5	0.45 λ_0× 0.41 λ_0	0.03 λ_0	Yes	Omnidirectional
[20]	8.25	115.2 %	5.6	0.23 λ_0× 0.23 λ_0	0.12 λ_0	Yes	Directional
[21]	2.0	10 %	7	0.32 λ_0× 0.32 λ_0	0.10 λ_0	Yes	Directional
[22]	35.91	16.7 %	8.6	1.81 λ_0× 1.71 λ_0	1.81 λ_0	Yes	Directional
[23]	4.95	82.8 %	8.1	0.77 λ_0× 0.70 λ_0	0.12 λ_0	No	Directional
[31]	6.87	44 %	6.5	0.42λ_0× 0.42 λ_0	0.07 λ_0	No	Directional
[34]	6.59	41 %	6.6	0.48 λ_0 × 0.48 λ_0	0.07 λ_0	No	Directional
[47]	5.47	33 %	6.1	1.30 λ_0 × 1.30 λ_0	0.06 λ_0	No	Directional
[48]	5.20	42 %	7.9	1.0 λ_0 × 1.0 λ_0	0.08 λ_0	No	Directional
This work	6.87	64 %	6.9	0.68 λ_0 × 0.68 λ_0	0.07 λ_0	Yes	Directional

NM: Not Mention

incorporating tumors of different radii. A significant difference in reflection coefficients was observed between normal breast and breast models with tumors. Finally, the designed antenna sensor and the breast phantom are fabricated. The measured results of the sensor's gain, bandwidth, and radiation pattern show good agreement with the simulated results. The miniaturized sensor with an aperture dimension of 0.39λ0 × 0.51λ0 can be obtained. At the same time, the proposed sensor can obtain a wide bandwidth from 4.67 GHz to 9.07 GHz (64.0 %), a low profile of 0.07λ0, and a maximum gain of 6.9 dBi. Furthermore, the measured variations in reflection coefficients between normal and malignant phantoms further demonstrate the sensor's effectiveness for breast tumor detection.

The authors whose names are listed under the paper title "Miniaturized and Wideband Metasurface Antenna Sensor for Breast Tumor Detection" certify that they have no conflict of interest. This manuscript has not been published in whole or in part nor is it being considered for publication elsewhere.

CRediT authorship contribution statement

Le Chang: Writing – review & editing, Supervision, Software, Funding acquisition. **Yaozhong Cui:** Software, Methodology. **Zhenyu Zhao:** Writing – original draft, Validation, Software, Methodology, Investigation. **Anxue Zhang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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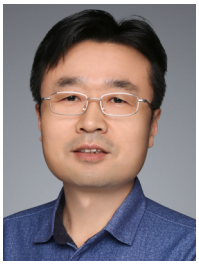
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